Audit Dashboard Analysis

Prepared By:
Nivetha M

Cyber Security Department

Sri Krishna College of Technology

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1. Executive Summary

This report presents the findings of a **data-driven financial audit** conducted using a simulated dataset and an interactive dashboard. The objective was to replicate a real-world audit environment and evaluate transactional integrity, compliance, and risk exposure.

The analysis identified **multiple high-risk transactions, duplicate entries, and control inefficiencies** across departments. The dashboard-enabled approach allowed for **real-time anomaly detection and data visualization**, improving audit effectiveness.

Key Highlights:

- High-risk transactions detected across multiple departments
- Duplicate transactions identified, indicating potential control gaps
- Presence of unapproved and delayed transactions
- Concentration of anomalies in Finance and IT departments

2. Audit Objectives

The primary objectives of this audit engagement were:

- To assess the integrity and reliability of transaction data
- To identify anomalies, fraud indicators, and compliance issues

- To evaluate the effectiveness of internal controls
- To provide actionable insights through data visualization
- To simulate a real-time auditing environment using dashboards

3. Scope and Coverage

The audit covers transactional data across the following functional areas:

- Finance
- Human Resources (HR)
- Information Technology (IT)
- Sales
- Procurement

Dataset Attributes:

The dataset includes structured financial data with fields such as:

- Transaction ID and Date
- Department and Vendor
- Employee ID
- Transaction Amount (INR)
- Payment Method
- Invoice Number
- Approval Status
- Risk Category
- Audit Flags and Indicators
- Duplicate and Late Payment Markers

4. Audit Methodology

The audit was conducted using a **data analytics-driven approach**, comprising the following stages:

4.1 Data Preparation

- Data validation and cleansing
- Standardization of formats
- Removal of inconsistencies

4.2 Analytical Procedures

- Trend analysis across time periods
- Department-wise transaction segmentation
- Risk categorization based on predefined rules
- Identification of outliers and anomalies

4.3 Control Testing

- Verification of approval workflows
- Duplicate transaction checks
- Late payment identification
- High-value transaction scrutiny

4.4 Visualization & Reporting

An interactive dashboard was developed to:

- Present KPIs
- Enable filtering and drill-down analysis
- Highlight high-risk and flagged transactions

5. Dashboard Overview

The audit dashboard serves as a **decision-support system** for auditors.

Key Components:

- **KPI Metrics:** Total Transactions, Total Amount, High-Risk Count, Flagged Entries
- **Risk Distribution Analysis:** Visual breakdown of risk categories
- **Departmental Insights:** Transaction patterns across departments
- **Time-Series Analysis:** Trends over time
- **Detailed Transaction Table:** Focus on flagged and anomalous entries

User Interactivity:

- Dynamic filters (Date, Department, Risk Level, Location)
- Drill-down capabilities for detailed inspection

6. Detailed Audit Findings

6.1 High-Risk Transactions

A notable proportion of transactions were classified as **high-risk**, primarily due to:

- High transaction amounts
- Missing or pending approvals
- Suspicious payment patterns

These transactions require **immediate audit attention**.

6.2 Duplicate Transactions

Duplicate entries were identified, especially in the **Finance department**, suggesting:

- Weak validation controls
- Potential data entry duplication

6.3 Approval Process Gaps

Transactions marked as **Pending or Rejected** were still present in financial records, indicating:

- Ineffective enforcement of approval workflows
- Possible bypassing of internal controls

6.4 Late Payment Trends

Late payments were observed across departments, particularly in:

- IT
- Procurement

This may impact:

- Vendor relationships
- Organizational credibility
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6.5 High-Value Transactions

Certain transactions exceeded expected thresholds and were flagged as **potential outliers**, requiring further verification.

7. Risk Assessment

Transactions were categorized into three risk levels:

Risk Level	Description
Low	Fully compliant, low probability of error
Medium	Requires monitoring and periodic review
High	Significant risk, requires immediate investigation

8. Recommendations

Based on the audit findings, the following actions are recommended:

8.1 Strengthen Internal Controls

- Implement automated validation checks
- Enforce strict approval hierarchies

8.2 Enhance Monitoring Mechanisms

- Continuous monitoring of high-risk transactions
- Real-time alerts for anomalies

8.3 Improve Data Governance

- Prevent duplicate entries through system constraints
- Ensure data accuracy and consistency

8.4 Optimize Payment Processes

- Implement timely payment tracking systems
- Reduce delays through automation

8.5 Adopt Advanced Analytics

- Use predictive analytics for fraud detection
- Integrate dashboards into regular audit workflows

9. Tools and Technologies

- **Data Visualization:** Power BI / Tableau / Streamlit
- **Data Processing:** Python (Pandas) / Excel
- **Dataset Format:** CSV

10. Limitations

- The dataset is simulated and may not fully reflect real-world complexities
- Limited sample size restricts large-scale inference
- Risk classification is based on predefined assumptions

11. Conclusion

This project demonstrates the effectiveness of **data analytics and visualization in modern auditing practices**.

The use of an interactive dashboard enables:

- Faster anomaly detection
- Improved transparency
- Data-driven decision-making

Adopting such tools in real-world auditing can significantly enhance **efficiency, accuracy, and risk management capabilities**.

12. Appendix

Sample Dataset:

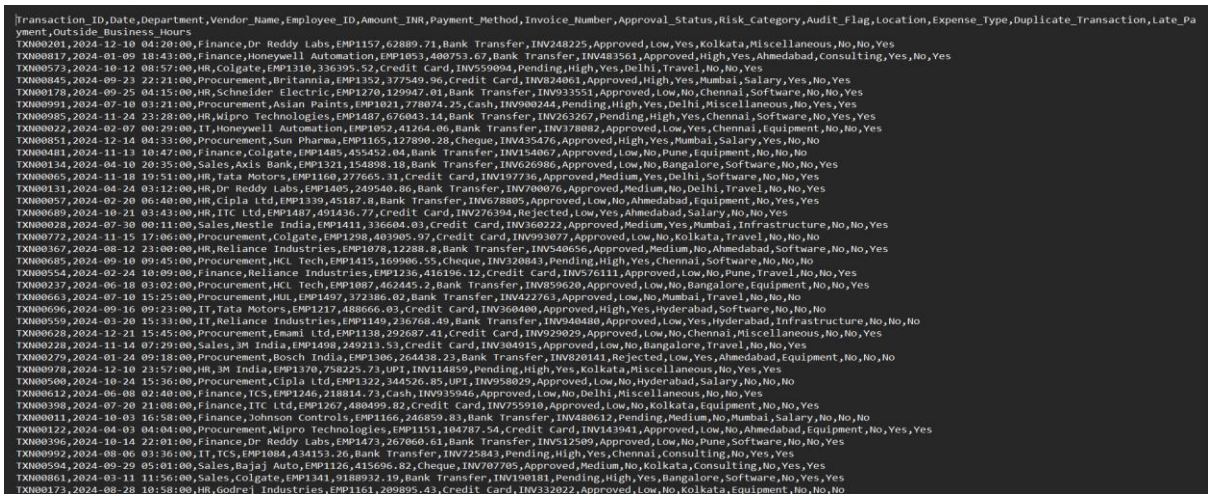

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Figure 12.1

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Figure 12.2

Dashboard:

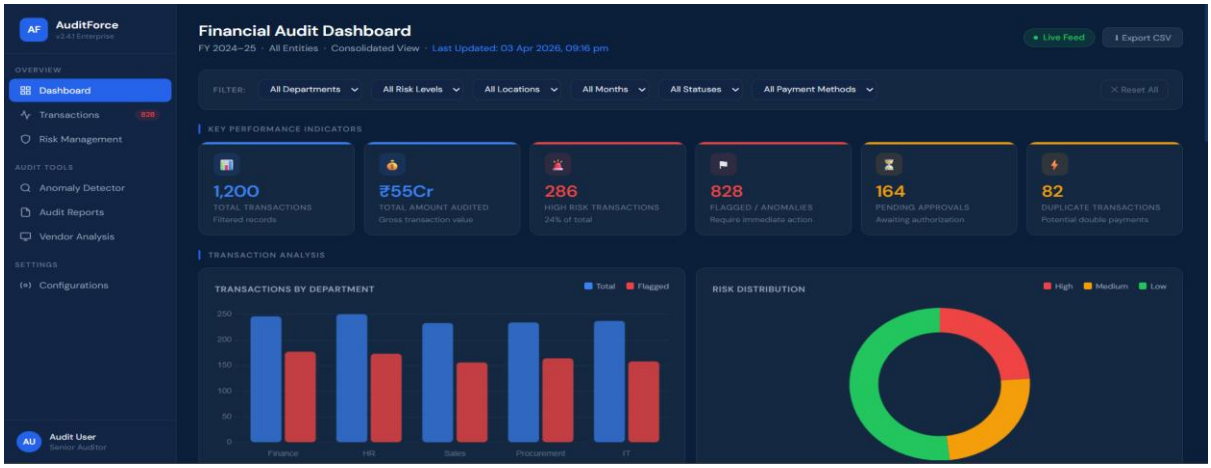

Figure 12.3

Figure 12.4

Figure 12.5

Figure 12.6